



DANCING STARS

A Binary Star System Research Project



Boyce-Astro

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PASS Summer Research Internship 2025

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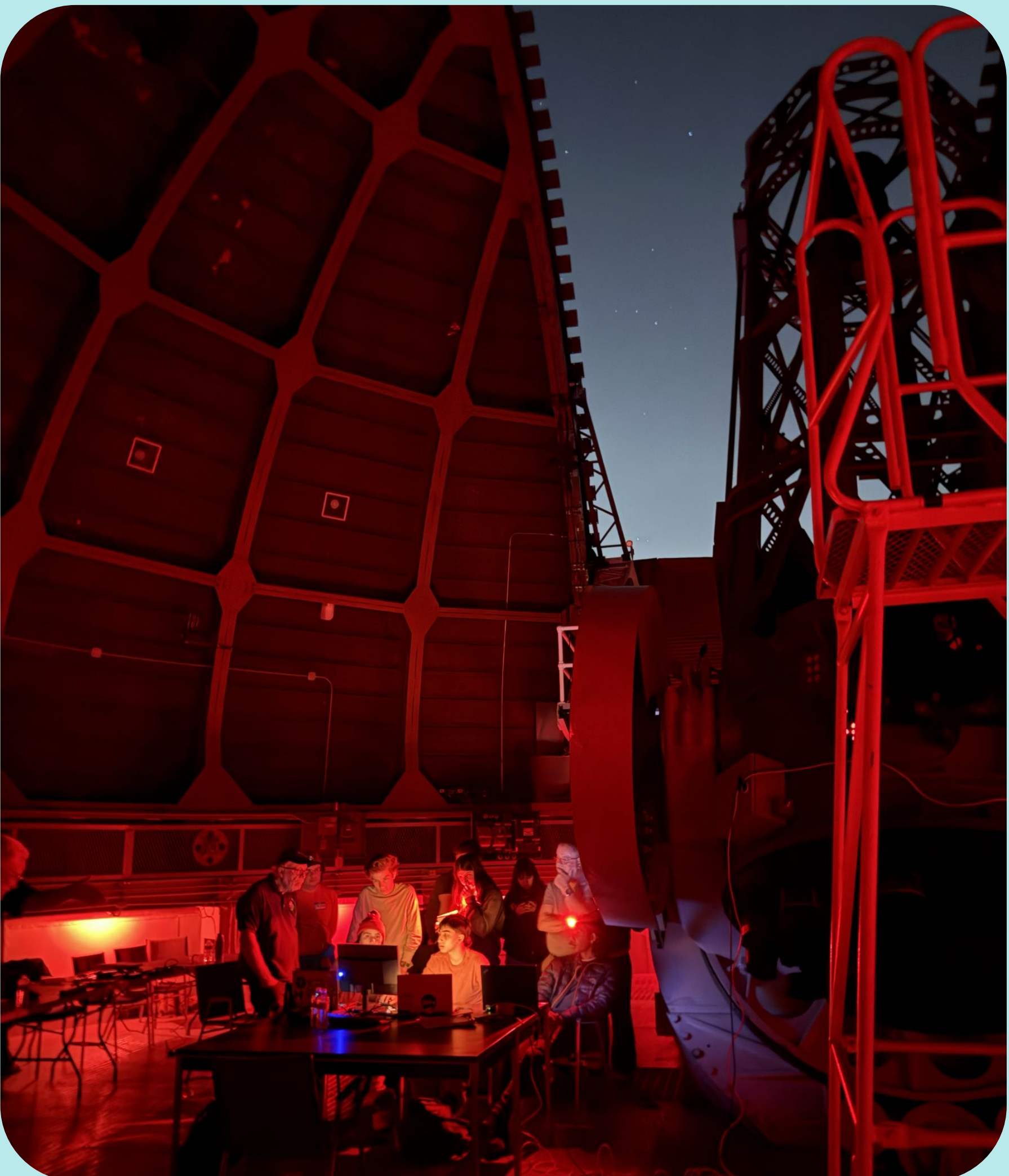
PURPOSE

A binary star system consists of two stars gravitationally bound and orbiting a common center of mass. This project aimed to measure the separation (ρ) and position angle (θ) of the binary system **WDS 17520+1520** to refine its orbital model.

OBSERVATION

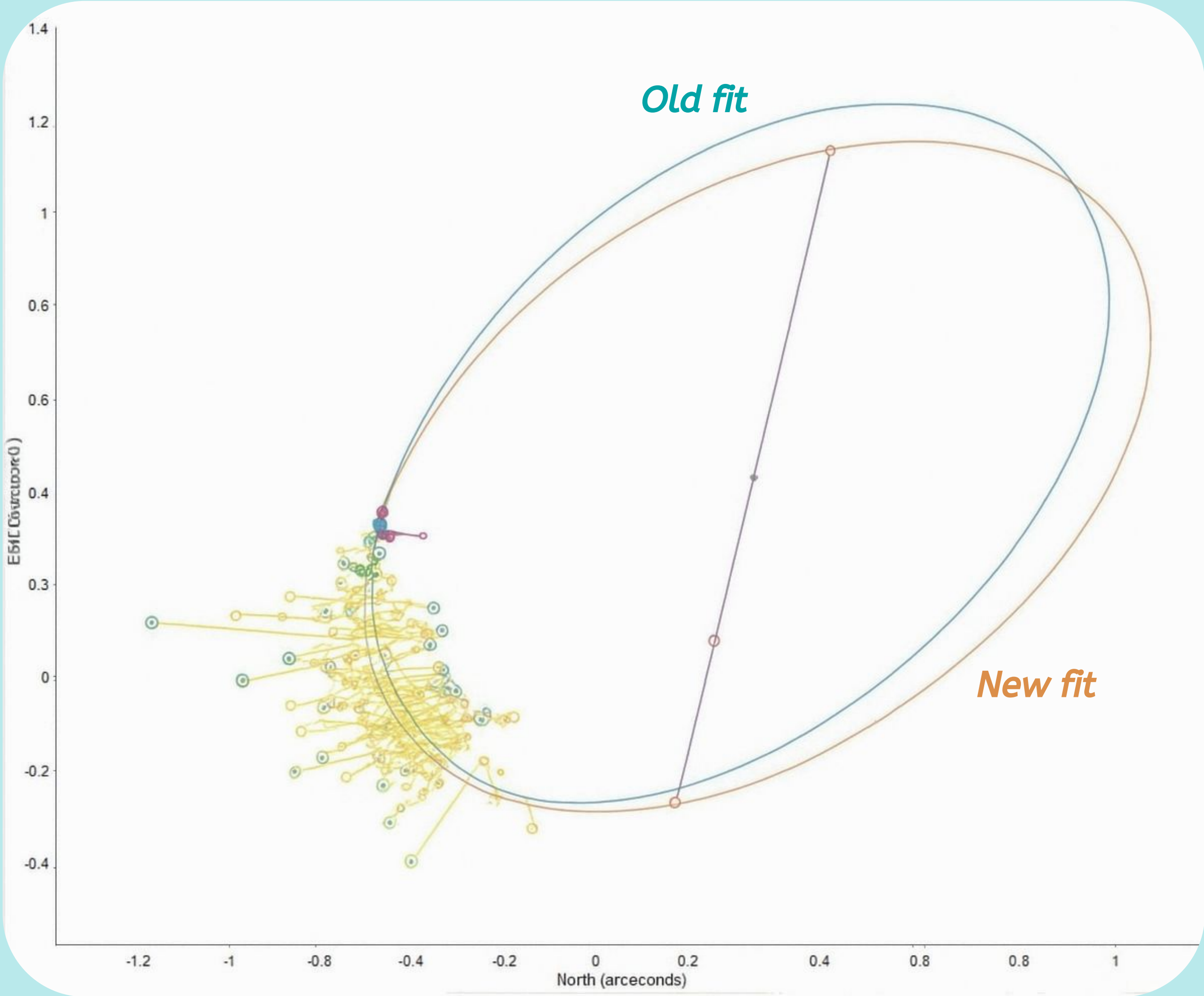
Mount Wilson Observatory – 60-inch Telescope

Aperture: 1.52 m
Pixel Scale: 0.1347" / px
Filter: Sloan R ($\lambda = 629$ nm)
Exposure: 0.08 s
Frames: 1000 (target), 300 (reference)
Reflections: 3



RESULTS

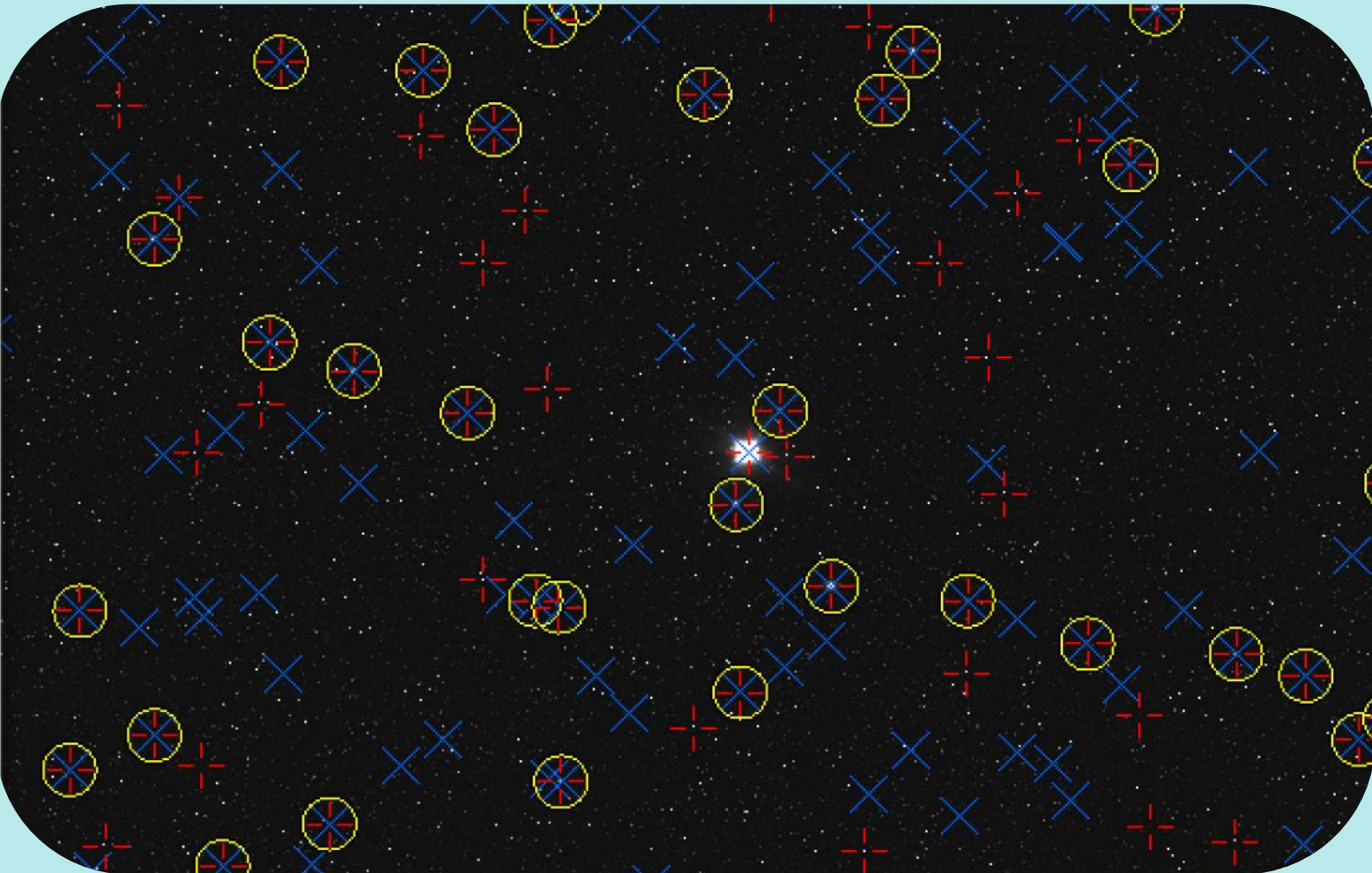
Measured values:
Separation (ρ) = 0.8359" $\pm$ 0.00091" (SEM)
Position Angle (θ) = 160.37° $\pm$ 0.13° (SEM)



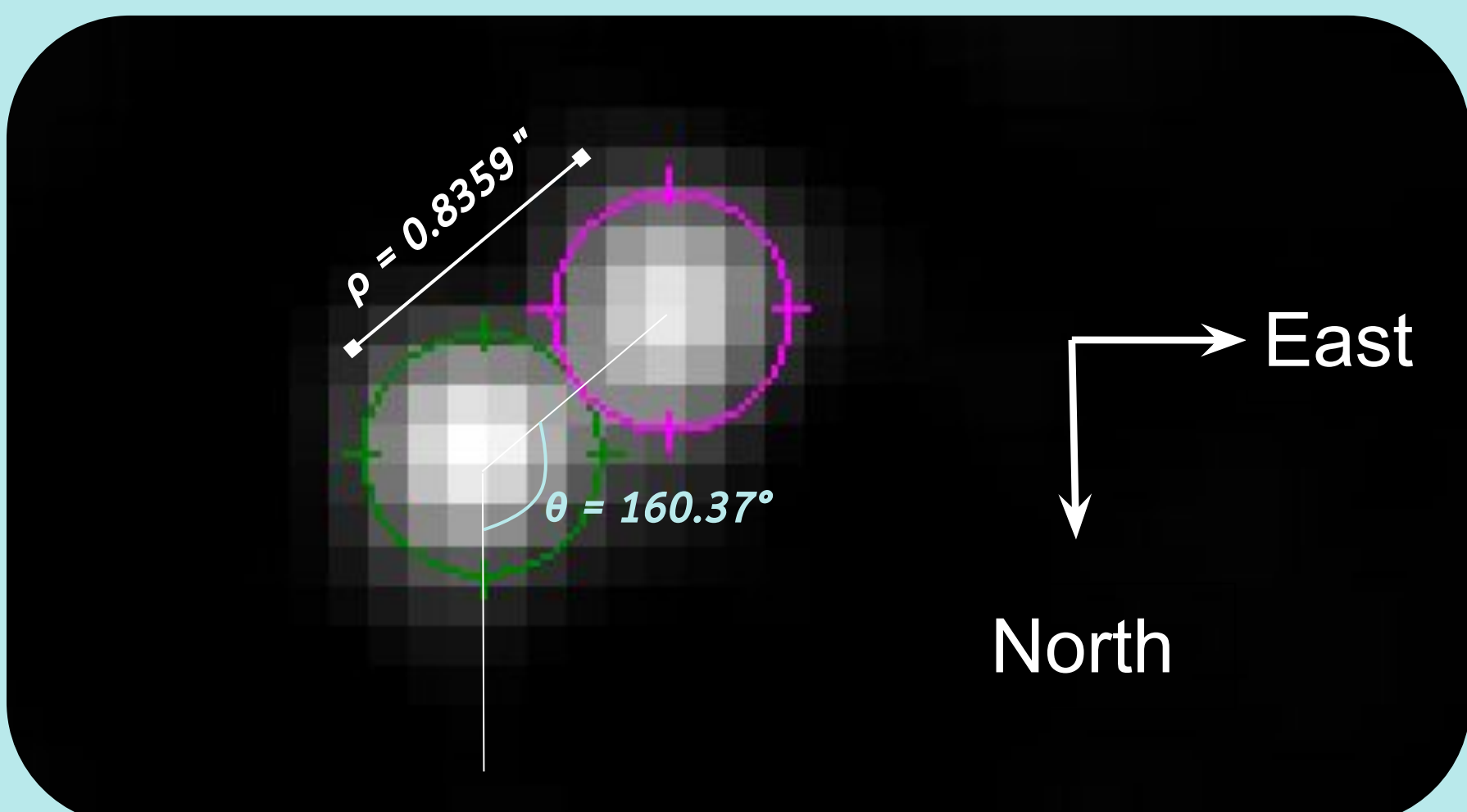
SYSTEM INFORMATION

- Constellation: Hercules
- Magnitudes: 7.21 and 7.38 ($\Delta m = 0.17$)
- Distance: 234 pc ($\approx$ 764 ly)
- Orbital Grade: 4 (1 = definitive)
- First Observed: 1843 | Last WDS Observation: 2021

SOFTWARE TOOLS



PlateSolve: determines pixel scale and camera orientation



Speckle Toolbox: reduces atmospheric distortion using Fourier analysis

CONCLUSION

Parameter	Old Model	New Model
Orbital Period (P)	1276.6 y	1267.8 y
Semi-major Axis (a)	1.140"	1.113"
Eccentricity (e)	0.582	0.518
Inclination (i)	124.2°	125.1°
WRMSD (Weighted Root Mean Square Deviation)	1367.90 mas	1338.58 mas

The updated fit exhibited a slightly shorter period, lower eccentricity, and reduced WRMSD, indicating an improved orbital solution. Future measurements are required to further constrain the long-period orbit.